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Prefactored optimized compact finite-difference schemes for second spatial derivatives

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ABSTRACT

We have described systematically the processes of developing prefactored optimized compact schemes for second spatial derivatives. First, instead of emphasizing high resolution of a single monochromatic wave, we focus on improving the representation of the compact finite difference schemes over a wide range of wavenumbers. This leads to the development of the optimized compact schemes whose coefficients will be determined by Fourier analysis and the least-squares optimization in the wavenumber domain. The resulted optimized compact schemes provide the maximum resolution in spatial directions for the simulation of wave propagations. However, solving for each spatial derivative using these compact schemes requires the inversion of a band matrix. To resolve this issue, we propose a prefactorization strategy that decomposes the original optimized compact scheme into forward and backward biased schemes, which can be solved explicitly. We achieve this by ensuring a property that the real numerical wavenumbers of both the forward and backward biased stencils are the same as that of the original central compact scheme, and their imaginary numerical wavenumbers have the same values but with opposite signs. This property guarantees that the original optimized compact scheme can be completely recovered after the summation of the forward and backward finite difference operators. These prefactored optimized compact schemes have smaller stencil sizes than even those of the original compact schemes, and hence, they can take full advantage of the computer caches without sacrificing their resolving power. Comparisons were made throughout with other well-known schemes.

INTRODUCTION

Reverse time migration (RTM) (Baysal et al., 1983; McMechan, 1983) directly utilizes a two-way scalar wave equation. Hence it provides a natural way to handle media with large velocity variations and does not suffer from the problem of imaging steep-dip reflectors that is inherent in one-way wave equation techniques. In the Gulf of Mexico, RTM has become a standard tool for imaging overturned reflections and other complicated structures. However, RTM generally requires a higher-order finite-difference scheme to approximate spatial derivatives that may consist of long stencils to reduce numerical grid dispersions (Dablain, 1986). This high computational demand prevents RTM from being applied practically for real data applications.

Finite difference is a technique that approximates the derivatives of a function at a point by using the function values at its neighboring points. Generally two main kinds of finite difference schemes can be used to approximate spatial derivatives: explicit schemes and compact or implicit schemes. To reach similar formal accuracy for the spatial derivatives at each grid point, explicit schemes have to utilize larger computational stencils, whereas the compact schemes utilize smaller stencils, but band (i.e., multidiagonal) matrices have to be solved implicitly.

The coefficients of the finite difference schemes were traditionally obtained by Taylor's theorem. However, it has been recognized recently that finite difference schemes that sacrifice formal order of accuracy, i.e., truncation errors, in favor of wide range of wavenumbers will provide significantly better wave propagation characteristics than traditional finite difference schemes (Vichnevetsky and Bowles, 1982). Based on this idea, people started to design explicit derivative operators by matching the spectral response of the operators in the wavenumber domain (Holberg, 1987; Tam and Webb,

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1993). This approach has improved the capability to propagate waves with shorter wavelengths over the traditional finite difference method. However, these spectral-optimal explicit finite differences still fail to bring significant improvement in accuracy, even with large computational stencils. This is the so-called "saturation effect" as described by Kosloff et al. (2010). In other words, although higher-order explicit schemes improve the accuracy, the rate of improvement for the finite-difference operator decreases.

Unlike explicit finite difference operators that are obtained by power series expansion, the spatial derivative operators can be approximated by rational expansions that are known to converge faster than power series expansion (Lele, 1992; Zhou et al., 2008; Kosloff et al., 2010). These implicit or recursive derivative operators, which have to obtain the solutions from inversion of either pentadiagonal or tridiagonal matrices, can efficiently improve the spatial resolution and the formal truncation errors. In fact, Lele (1992) has presented the pentadiagonal compact scheme for evaluating spatial derivatives. The scheme has the characteristics of the fourth-order of formal accuracy, but with a spectral-like resolution. He demonstrated that spectral-like resolution can be achieved by fitting the modified wavenumbers to the corresponding true wavenumbers to obtain the optimized coefficients for the compact schemes.

In this paper, we present a group of optimized compact finite-difference schemes that have a fourth-order of formal accuracy for a second spatial derivative, but with higher resolving power over a wide range of wavenumbers compared especially to explicit finite difference schemes. Although these compact schemes are more accurate than traditional explicit schemes with the same order of formal accuracy, solving for each spatial derivative requires inverting a band matrix that can be either a tridiagonal or a pentadiagonal matrix. Following the approach of Ashcroft and Zhang (2003), we propose a prefactorization strategy that will split the original band matrix from the compact schemes into two independent upper and lower diagonal matrices that are easier to solve. Therefore, instead of solving the band matrices using traditional LU decomposition, we solve them using a $L+U$ prefactorization method. With such an approach, a new class of highly accurate prefactored compact schemes is derived that uses even shorter stencils with coefficients optimized for a wide band of wavenumbers. Because of the application of shorter stencils, the data for finite difference schemes has great potential to fit into computer caches and at the same time improve the resolution power of the scheme. Fourier analyses have been used extensively in the paper to help construct and analyze the proposed prefactorized, optimized, compact finite difference schemes. A couple of synthetic examples have been demonstrated to illustrate the correctness and effectiveness of the algorithms.

THEORY

Seismic wave propagations are often governed by a 3D scalar wave equation, which is represented by

$$\frac{1}{v^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} + s(x,y,z,t), \tag{1}$$

where $u(x,y,z,t)$ denotes the wavefield, $v(x,y,z)$ is the velocity field for seismic wave propagation, and $s(x,y,z,t)$ is the source term. The dispersion relation for wave equation 1 can be obtained by Fourier analysis and can be described by:

$$\omega^2 = v^2 k^2. \tag{2}$$

Here, $k^2 = k_x^2 + k_y^2 + k_z^2$, in which k_x, k_y, and k_z are wavenumbers along each corresponding axis, respectively, and $\omega = 2\pi f$ is the angular frequency with f as the frequency. The phase velocity is the velocity with which a monochromatic wave with wavenumber k propagates and is given by

$$v_p = \frac{\omega}{|k|}. \tag{3}$$

For scalar wave equation 1, we have $v_p = v$, which implies that the original wave equation is dispersion-free. Without loss of generality, we will only consider a semidiscretization in space to the scalar wave equation. In other words, we assume that there are no errors in the time integration for equation 1. This can be achieved in practice by use of either very small time steps or a time integration scheme with higher order of accuracy.

Description of compact schemes

In the following, we will only discuss second derivative $\partial^2 u/\partial x^2$ in equation 1. Derivatives for other space dimensions can be similarly obtained. According to Lele (1992), a general compact approximation to the second spatial derivative $\partial^2 u/\partial x^2$ may be written in the form:

$$\begin{aligned}\beta D_{i-2}^2 + \alpha D_{i-1}^2 + D_i^2 + \alpha D_{i+1}^2 + \beta D_{i+2}^2 \\ = c\frac{u_{i+3} - 2u_i + u_{i-3}}{9h^2} + b\frac{u_{i+2} - 2u_i + u_{i-2}}{4h^2} \\ + a\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2},\end{aligned} \tag{4}$$

where D_i^2 is the second spatial derivative of the function u at grid location x_i and h is the grid space interval. The relations among the coefficients a, b, c, α, and β in the compact scheme 4 can be obtained by matching the Taylor series coefficients of the specified orders of formal accuracy. The truncation error of the approximation is determined by the first unmatched coefficients. These relations are:

$$a + b + c = 1 + 2\alpha + 2\beta \qquad \text{(second-order)}, \tag{5}$$

$$a + 2^2 b + 3^2 c = \frac{4!}{2!}(\alpha + 2^2\beta) \qquad \text{(fourth-order)}, \tag{6}$$

$$a + 2^4 b + 3^4 c = \frac{6!}{4!}(\alpha + 2^4\beta) \qquad \text{(sixth-order)}, \tag{7}$$

$$a + 2^6 b + 3^6 c = \frac{8!}{6!}(\alpha + 2^6\beta) \qquad \text{(eighth-order)}, \tag{8}$$

$$a + 2^8 b + 3^8 c = \frac{10!}{8!}(\alpha + 2^8\beta) \qquad \text{(tenth-order)}. \tag{9}$$

From these equations, it is easy to see that only the eighth-order tridiagonal ($\beta = 0$) and the tenth-order pentadiagonal ($\beta \neq 0$) schemes have unique coefficients. The other lower-order schemes have free coefficients that are not determined until more constraints are imposed or the stencil size is reduced. These free coefficients provide the freedom for further optimization of the schemes.

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Accuracy of the generic compact scheme 4 can be measured from a Fourier analysis of the spatial derivative. For this purpose, we let $w = kh$ be the modified or scaled wavenumber. Obviously, from Fourier analysis, $w^2 = k^2h^2$ is the (scaled) true wavenumber for the second derivative. Similarly, the numerical approximation $w''(w)$ of second derivative can be found by applying Fourier analysis to equation 4:

$$w''(w) = \frac{2a[1-\cos(w)] + \frac{b}{2}[1-\cos(2w)] + \frac{2c}{9}[1-\cos(3w)]}{1+2\alpha\cos(w)+2\beta\cos(2w)} = \frac{\left[2a+\frac{b}{2}+\frac{2c}{9}\right] - 2a\cos(w) - \frac{b}{2}\cos(2w) - \frac{2c}{9}\cos(3w)}{1+2\alpha\cos(w)+2\beta\cos(2w)}. \quad (10)$$

Here, $w''(w) = \tilde{k}^2h^2$ is called effective (scaled) wavenumber for the second derivative. It is a periodic function of $w = kh$ (the scaled wavenumber) with a period of 2π. The difference between $w''(w)$ and w^2 measures the approximation error of the second derivative and will be used for the analysis of the accuracy of finite difference schemes. To ensure that the Fourier transform of the finite-difference scheme is a good approximation of the partial derivative, the effective wavenumber $w''(w)$ should coincide with the corresponding (scaled) true wavenumber w^2 over as wide a range of wavenumbers (e.g., $0 \le w < \pi$) as possible. The effectiveness of a finite difference scheme can also be examined by considering phase velocity as a function of frequency or, equivalently, as a function of G, points-per-wavelength (PPW). See Appendix A for details.

Optimization of coefficients

Finite difference schemes that sacrifice formal order of accuracy in favor of wide-band performance can provide significantly better wave propagation characteristics in the high wavenumber range. Under our assumption of semidiscretization, minimizing the error on the effective wavenumber is equivalent to controlling the phase velocity error that is relevant for a simple harmonic plane wave. Therefore, optimized schemes can be obtained by minimizing the difference between the effective wavenumber $w''(w)$ and the true wavenumber w^2 which is defined as an integral error

$$\min_{\eta_1,\ldots,\eta_3} E(\eta_1,\eta_2,\eta_3,\eta_4,\eta_5) = \int_{w_l}^{w_u} |w'' - w^2|^2 \xi(w)\mathrm{d}w, \quad (11)$$

over a large wavenumber range $0 \le w_l = k_l h \le w = kh \le w_u = k_u h < \pi$. Here $\eta_i(\eta_1 = a, \eta_2 = b, \eta_3 = c, \eta_4 = \alpha, \eta_5 = \beta)$ correspond to the coefficients of the basic stencil as defined by equation 4. And $\xi(w)$ is the weight function for the integrals that will be used to boost or reduce certain range of wavenumbers. According to Kim and Lee (1996), a carefully chosen weight function $\xi(w)$ can both make the equation 11 analytically integrable and allow the integrand to be weighted to the wide wavenumber range. For our case study, the weighting function has the form

$$\xi(w) = [1 + 2\alpha\cos(w) + 2\beta\cos(2w)]^2. \quad (12)$$

It is necessary to find the optimum values of the coefficients that minimize the error in equation 11. Generally, minimization of equation 11 could be obtained by setting

$$\frac{\partial E}{\partial \eta_i} = 0, \qquad i = 1, 2, \ldots, 5. \quad (13)$$

This provides a system of linear algebraic equations by which the coefficients $\eta_i(\eta_1 = a, \eta_2 = b, \eta_3 = c, \eta_4 = \alpha, \eta_5 = \beta)$ can be easily determined.

Although an optimized lower-order compact scheme may have more resolving power in the wavenumber domain than an optimized higher-order one, it is expected that the lower-order scheme may produce more truncation errors (actual errors in space) than the higher-order one (Jastram and Behle, 1993; Kim and Lee, 1996). The overall accuracy of a finite difference scheme should be determined by not only the resolution in the wavenumber domain but also by the truncation order in space for actual computations. Those compact schemes that have the best combinations of the resolution in wavenumber domain and the order of truncation errors appear to be the most effective. The range of wavenumbers $[w_l, w_u]$ over which the effective wavenumber approximates the exact derivative wavenumber will define a set of well-resolved waves too. Generally, w_l is chosen to be around zero. The value of the cutoff wavenumber w_u, which gives the shortest well-resolved wave, depends on the specified error tolerance. In our case, we chose $w_u = 0.9\pi$ so that the square root of the accumulated errors E in equation 11 is around 0.5%. It is necessary to constrain the overshoot (the largest deviation from the exact derivative wavenumber in a given range) within a tolerable range. For our study, we choose to maintain at least fourth-order formal accuracy by satisfying both equations 5 and 6. Combining three extra equations from equation 13, we obtain a system of linear algebraic equations by which the optimal coefficients can be derived:

$$\alpha = 0.505737, \quad \beta = 0.0553883, \quad a = 0.203893,$$
$$b = 1.74833, \quad c = 0.170032. \quad (14)$$

This results in a pentadiagonal optimized compact scheme that has the fourth-order of formal accuracy but a spectral-like resolution.

Accuracy of compact schemes

For comparisons, we have selected four finite difference schemes: the simplest compact scheme as proposed by Claerbout (1985) (also see Appendix B for details), the eighth-order explicit finite difference scheme, the tridiagonal optimized compact scheme (see Appendix B for details), and the pentadiagonal optimized compact scheme. From now on, we will describe them as scheme A, scheme B, scheme C, and scheme D, respectively.

The difference between $w''(w)$ and w^2 is a measure of the approximation error of the second spatial derivative. Figure 1 shows plots of the effective wavenumbers $w''(w)$ against the modified or scaled wavenumber w for a variety of schemes mentioned above. It can be found that, compared to other standard central differences, our optimized compact schemes, both pentadiagonal (scheme D) and tridiagonal (scheme C), stay close to the exact differentiation over a wider range of wavenumbers. Figure 2 shows the relative errors of the effective wavenumber with respect to G (PPW) for these finite difference schemes. Obviously, our proposed optimized schemes need fewer G. The dispersive error characteristics can be alternatively presented in terms of the error in the phase speed of

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waves of different wavenumber. Ideally, the normalized phase velocity of the operator should be constantly one for all wavenumber components. Figure 3 shows the 1D normalized phase velocity for the above set of schemes (see equation A-5). Once again the improved phase error of the optimized compact schemes is evident. It shows that schemes with wide-band wavenumbers can be obtained by not insisting on the highest possible formal accuracy. In 2D or 3D problems, the phase errors of the finite difference schemes also appear in the form of numerical anisotropy. The numerical anisotropic propagation is displayed in Figure 4 for these finite-difference schemes (see equation A-6). It is interesting to note that high-resolution finite difference schemes also help to reduce numerical anisotropy.

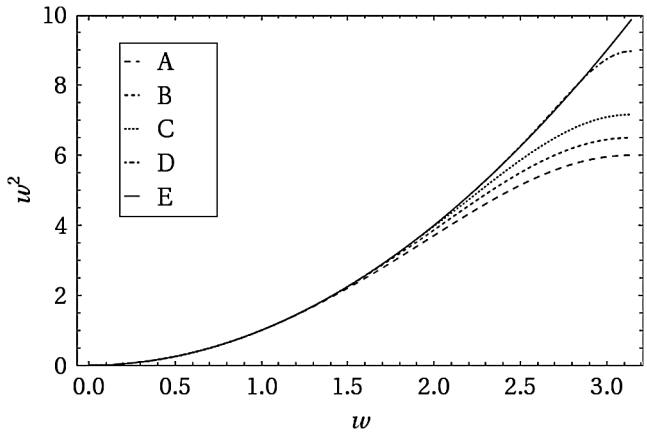

Figure 1. Dispersion relations of different finite difference approximations of second derivative: Curve A corresponds to scheme with Claerbout's "1/10" trick (scheme A), curve B corresponds to standard eighth-order explicit finite difference scheme (scheme B), curve C corresponds to an optimized tridiagonal compact scheme (scheme C), curve D corresponds to an optimized pentadiagonal compact scheme (scheme D), and curve E is the exact solution. Note that our proposed scheme D is very close to the true wavenumber.

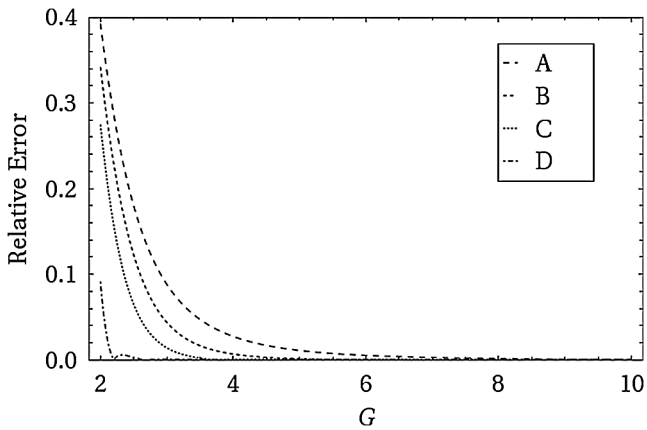

Figure 2. Relative phase errors of different schemes with respect to the number of grid PPW G. The curves correspond to Schemes A, B, C, D from the outermost to the innermost, respectively.

Prefactored compact schemes

While compact schemes are more accurate than the equivalent explicit schemes, it is known that they have one practical drawback: A band matrix must be inverted to obtain the spatial derivative at a grid point. In this paper, we will present a strategy to prefactorize these compact schemes so as to obtain the solutions without matrix inversion. For simplicity, we consider in the following only tridiagonal compact schemes ($\beta = 0$) in equation 4. However, application of the analysis to pentadiagonal compact schemes should be straightforward. To derive the prefactorized compact schemes, we follow Ashcroft and Zhang (2003) by defining the second derivative D_i^2 as the average of the forward and backward spatial derivative operators $D_i^{2,F}$ and $D_i^{2,B}$ such that

$$D_i^2 = \frac{1}{2}(D_i^{2,F} + D_i^{2,B}), \tag{15}$$

where the biased stencils for operators $D_i^{2,F}$ and $D_i^{2,B}$ are defined, respectively as follows

$$\alpha_F D_{i+1}^{2,F} + \beta_F D_i^{2,F} = \frac{1}{h^2}[a_F u_{i+2} + b_F u_{i+1} + c_F u_i + d_F u_{i-1} + e_F u_{i-2}], \tag{16}$$

$$\beta_B D_i^{2,B} + \gamma_B D_{i-1}^{2,B} = \frac{1}{h^2}[a_B u_{i+2} + b_B u_{i+1} + c_B u_i + d_B u_{i-1} + e_B u_{i-2}]. \tag{17}$$

This is equivalent to splitting the original implicit tridiagonal matrix into two independent upper and lower bidiagonal matrices. Unlike the original (nondissipative) compact schemes, which are symmetric and hence their numerical wavenumber will contain only real numbers, the forward and backward operators in equations 16 and 17 represent biased (dissipative) stencils, which means that their numerical wavenumbers will contain both real and imaginary components. The coefficients are chosen such that the original compact scheme can be completely recovered when the two generic biased

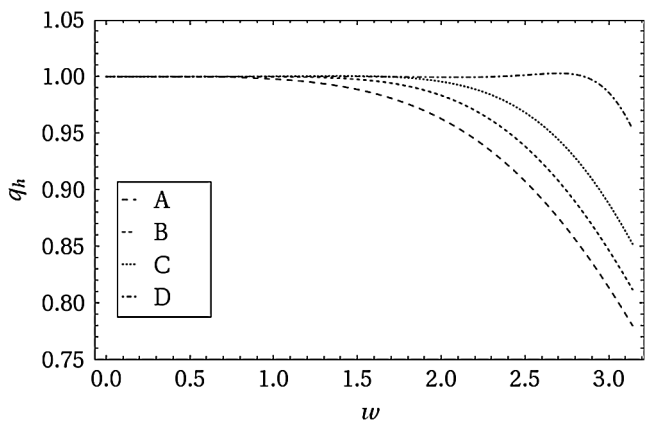

Figure 3. Normalized phase velocity for different schemes. Curve A corresponds to scheme with Claerbout's "1/10" trick (scheme A), curve B corresponds to standard eighth-order explicit finite-difference scheme (scheme B), curve C corresponds to an optimized tridiagonal compact scheme (scheme C), curve D corresponds to an optimized pentadiagonal compact scheme (scheme D).

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stencils in equations 16 and 17 are summed together. This can be achieved by a property that is revealed by Hixon and Turkel (2000) during the analysis of their compact MacCormack schemes. It is sufficient to ensure that the real numerical wavenumbers of the forward and backward biased stencils are the same as those of the original central compact scheme, and their imaginary numerical wavenumbers have the same absolute values but opposite signs. In this way, after summing the forward and backward operators, the original compact scheme can be recovered, while the imaginary numerical wavenumber components of the forward and backward operators will cancel each other. Although they discussed this property for first derivatives, it is also valid for second derivatives.

To obtain the numerical wavenumber of the generic forward and backward operators for second spatial derivatives, we will

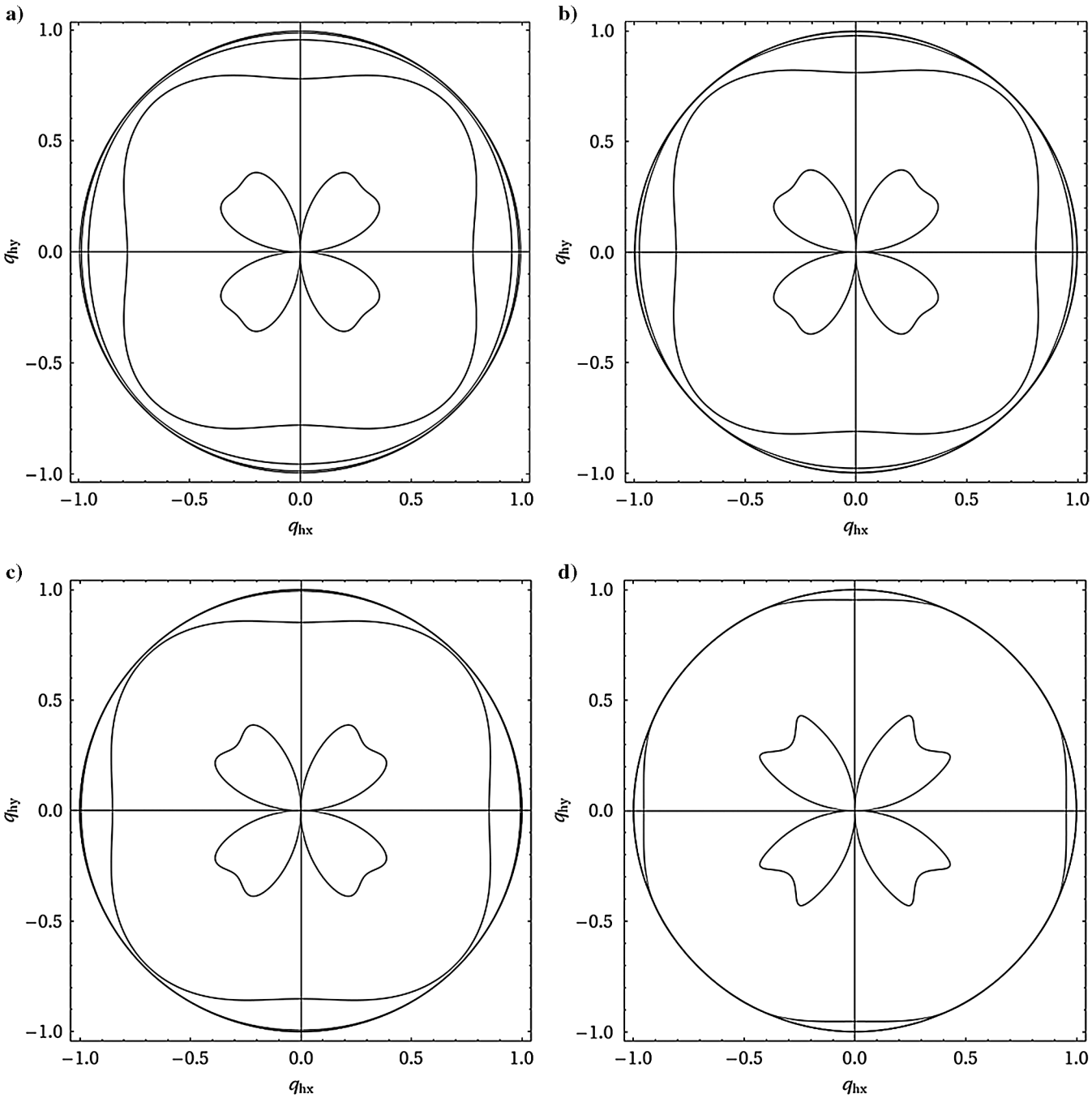

Figure 4. Polar plots of normalized phase speed anisotropy for various schemes. The phase speeds for PPW G=1, 2, 3, 4, 5 are plotted from innermost to the outermost, respectively. (a) Scheme A (Claerbout's simplest compact scheme); (b) scheme B (standard eighth-order explicit finite-difference scheme); (c) scheme C (optimized fourth-order tridiagonal compact scheme); and (d) scheme D (optimized fourth-order pentadiagonal compact scheme).

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apply a Fourier analysis similar to the original compact scheme 4. Therefore, for the forward stencil defined by equation 16, the real and imaginary components of the numerical wavenumber are given separately by

$$Re(w_F'') = \frac{(b_F\alpha_F + c_F\beta_F) + (a_F\alpha_F + b_F\beta_F + c_F\alpha_F + d_F\beta_F)\cos(w)}{\alpha_F^2 + \beta_F^2 + 2\alpha_F\beta_F\cos(w)} + \frac{(a_F\beta_F + d_F\alpha_F + e_F\beta_F)\cos(2w) + e_F\alpha_F\cos(3w)}{\alpha_F^2 + \beta_F^2 + 2\alpha_F\beta_F\cos(w)}, \quad (18)$$

$$Im(w_F'') = \frac{(-a_F\alpha_F - b_F\beta_F + c_F\alpha_F + d_F\beta_F)\sin(w)}{\alpha_F^2 + \beta_F^2 + 2\alpha_F\beta_F\cos(w)} + \frac{(-a_F\beta_F + d_F\alpha_F + e_F\beta_F)\sin(2w) + e_F\alpha_F\sin(3w)}{\alpha_F^2 + \beta_F^2 + 2\alpha_F\beta_F\cos(w)}. \quad (19)$$

Here, $w_F'' = \tilde{k}_F^2 h^2$ is the effective wavenumber for the forward operator. For the backward stencil defined by equation 17, similar results can be obtained

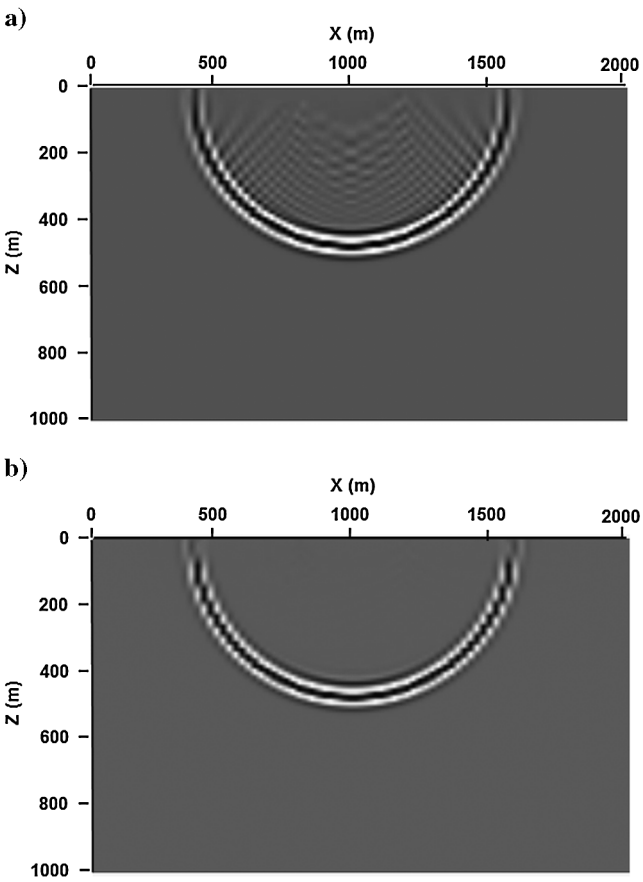

Figure 5. Impulse responses for prestack RTM. (a) Extrapolated with a standard eighth-order explicit finite-difference scheme (scheme B). Note the strong artifacts that are caused by numerical grid dispersions. (b) Extrapolated with an optimized pentadiagonal compact scheme (scheme D). Note the much weaker numerical grid dispersion.

$$Re(W_B'') = \frac{(c_B\beta_B + d_F\gamma_B) + (b_B\beta_B + c_B\gamma_B + d_B\beta_B + e_B\gamma_B)\cos(w)}{\beta_B^2 + \gamma_B^2 + 2\beta_B\gamma_B\cos(w)} + \frac{(a_B\beta_B + b_B\gamma_B + e_B\beta_B)\cos(2w) + a_B\gamma_B\cos(3w)}{\beta_B^2 + \gamma_B^2 + 2\beta_B\gamma_B\cos(w)}, \quad (20)$$

$$Im(w_B'') = \frac{(-b_B\beta_B - c_B\gamma_B + d_B\beta_B + e_B\gamma_B)\sin(w)}{\beta_B^2 + \gamma_B^2 + 2\beta_B\gamma_B\cos(w)} + \frac{(-a_B\beta_B - b_B\gamma_B + e_B\beta_B)\sin(2w) - a_B\gamma_B\sin(3w)}{\beta_B^2 + \gamma_B^2 + 2\beta_B\gamma_B\cos(w)} \quad (21)$$

Here $w_B'' = \tilde{k}_B^2 h^2$ is the effective wavenumber for the backward operator.

Many combinations of coefficients will satisfy equations 18, 19, 20, and 21. As one possibility, the coefficients of the backward stencil can be first chosen as follows to ensure that the imaginary components of forward and backward operators have the same absolute values but opposite signs

$$\beta_B = \beta_F, \quad \gamma_B = \alpha_F, a_B = e_F, \quad b_B = d_F,$$
$$c_B = c_F, \quad d_B = b_F, \quad e_B = a_F. \quad (22)$$

Then matching various terms in equation 10 with those in equation 18 will provide the coefficients of the forward stencil. The procedures of prefactorization of a compact scheme are demonstrated in Appendix B.

NUMERICAL EXAMPLES

In this section we will compare the effectiveness of the prefactored fourth-order optimized compact finite-difference scheme (scheme D) with that of traditional explicit eighth-order explicit finite-difference scheme (scheme B). Two synthetic examples are being used for the tests. We use second-order finite-difference scheme to approximate the time derivative in scalar wave equation 1. To insure that the discretization in time direction will not introduce visible numerical dispersions, we have used a small extrapolation time step in both tests. For example, we chose the extrapolation time step for the first test as 0.5 ms.

The first test will compare the results of 3D impulse responses for prestack RTM computed with both schemes. Figure 5a and 5b shows the vertical slices of the impulse responses of 3D prestack RTM depth migration computed with scheme B and scheme D, respectively. The horizontal grid spacings are defined as: $\Delta x = \Delta y = 20$ m, and the vertical spacing is $\Delta z = 10$ m. The model has a constant velocity of 2000 m/s. The source wavelet is Ricker with peak frequency of 30 Hz. The source and receiver locations are located at two sides with equal distance from the middle of the surface locations. Because of the larger grid spacings and the high frequency, the result in Figure 5a that was computed with standard eighth-order explicit finite-difference scheme shows strong numerical artifacts, while the result in Figure 5b that was obtained with our optimized pentadiagonal compact scheme is relatively clean of numerical dispersions.

The second test we have performed is the numerical simulation of the wave propagations in Sigsbee 2A model. Figure 6 shows the 2D Sigsbee 2A stratigraphic velocity model. It has 3201 points in horizontal direction and 1201 points in depth direction. The horizontal and depth grid sizes are both 25 ft. We embed a Ricker

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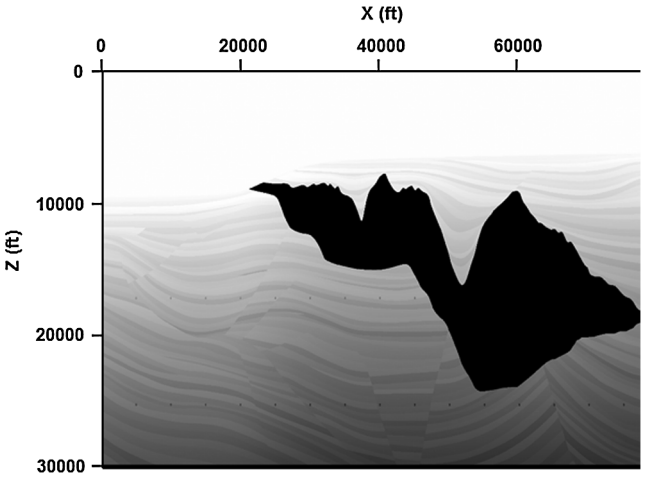

Figure 6. This is the Sigsbee 2A stratigraphic velocity model.

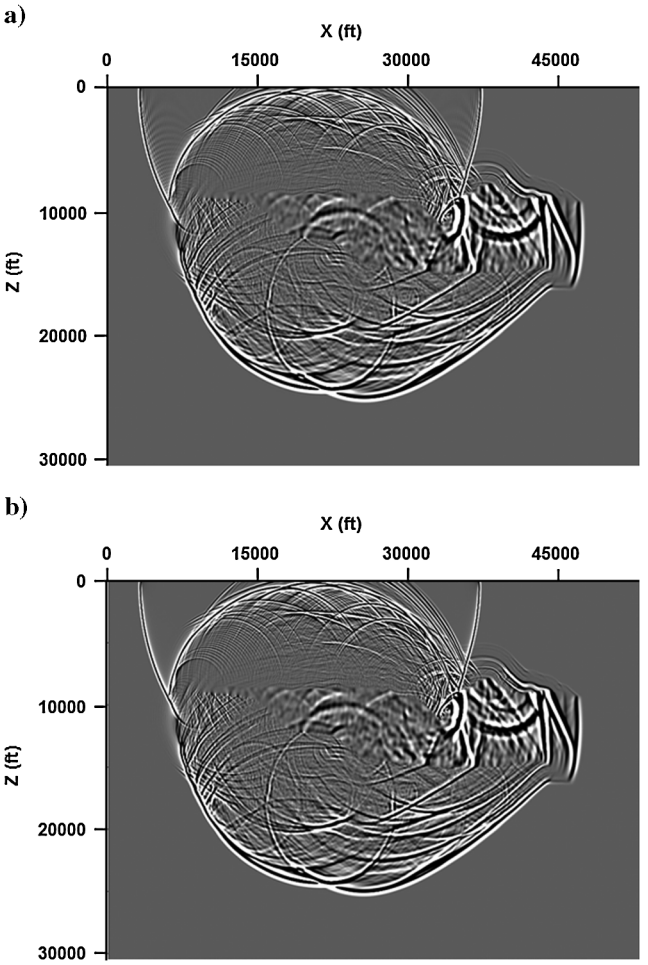

Figure 7. Snapshots of forward propagation wavefield at time $t = 3.45$ s. (a) This is computed with an eighth-order explicit finite-difference scheme (scheme B). Note the stronger numerical grid dispersion around the direct waves. (b) This is computed with a fourth-order pentadiagonal compact scheme (scheme D). Clearly, numerical grid dispersions around the direct waves have been greatly reduced.

source wavelet with dominant frequency of 20 Hz at the surface. The dimensions of the numerical grid for the computation are 1320 points in the horizontal direction and 1200 points in the vertical direction. The depth grid interval for the computation equals 25 ft and the lateral grid size equals 40 ft and they are fixed for both schemes. Figure 7a shows the snapshot of a forward propagation wavefield at time $t = 3.45$ s. This is computed with Scheme B. Note the strong numerical grid dispersions around the wavefronts of the direct wave. Numerical grid dispersions not only generate artificial waves, but also change the phases of the propagated waves. Therefore, although the numerical grid dispersions are not obvious for other events in Figure 7a, they exist and will propagate and interfere with other waves. Figure 7b is the snapshot of a forward propagation wavefield at the same time $t = 3.45$ s, but it is computed with scheme D. Clearly, numerical grid dispersions around the direct waves have been greatly reduced.

CONCLUSIONS

We mainly discuss and analyze the prefactored optimized compact schemes in 1D, but the method can be easily extended to 2D and 3D by applying the schemes along each of the spatial coordinates, separately. For 3D problems, we have implemented a hybrid technique that will apply a 2D compact scheme to the fastest two spatial dimensions, and an explicit optimized finite difference scheme to the slowest dimension for better performance. Compact finite difference scheme has the advantage of using shorter stencil lengths to approximate the spatial derivatives, but it also introduces complexities to the implementations of the algorithms. Issues of numerical stability and boundary conditions for compact schemes are among the active research topics in the past few years (Ashcroft and Zhang, 2003). Although we have not performed analysis on the stability conditions for the compact schemes, our educated guess is that its Courant-Friedrichs-Lewy (CFL) condition should be close to that of pseudospectral scheme because of their similar spectral responses (Kosloff et al., 2010). We use damping function similar to that of Cerjan et al. (1985) to attenuate the boundary reflections.

With the same computational configuration, i.e., same grid size and same time step, etc., the runtime for our prefactored optimized compact scheme should be in between the explicit finite-difference scheme and the pseudospectral algorithm. However, given the ability of using coarse grid, higher frequency, shorter stencil, and spectral-like resolving power, our prefactored compact scheme should perform superior to the eighth-order explicit finite-difference scheme and pseudospectral scheme.

We have first constructed a group of optimized compact finite difference schemes for second spatial derivative by Fourier analysis, and by applying the least-squares optimization technique to minimize the phase errors between the true and the numerical wavenumbers over a wide range of wavenumbers. We then propose a prefactorization strategy that will split the original band matrix that comes from the compact schemes into two independent upper and lower diagonal matrices that can be solved explicitly. These prefactorized compact schemes have shorter computational stencils, and therefore have better computational performance than that of original optimized compact schemes without sacrificing the resolving power.

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APPENDIX A

NORMALIZED PHASE-VELOCITY OF THE COMPACT SCHEMES

Expressions for phase velocity as a function of G, i.e., PPW, for the generic compact schemes of equation 4 will be developed in this appendix. Let's assume wavelength $\lambda = \frac{2\pi}{k} = \frac{v}{f}$, in which k is the wavenumber, f is the frequency, and v is the velocity of the original scalar wave equation 1. Therefore, G can be represented as $G = \frac{\lambda}{h} = \frac{2\pi}{w}$ (Alford et al., 1974). From equation 10, we obtain

$$w''(G) = \frac{2a[1-\cos(2\pi/G)] + \frac{b}{2}[1-\cos(4\pi/G)] + \frac{2c}{9}[1-\cos(6\pi/G)]}{1 + 2\alpha\cos(2\pi/G) + 2\beta\cos(4\pi/G)}. \tag{A-1}$$

This can be used to obtain the minimum grid points that the algorithm needs for the waves to propagate with less numerical dispersions. Similar to equation 2, the numerical dispersion relation for the semidiscretized compact schemes can be symbolically represented as follows

$$\omega_h^2 h^2 = v^2 w''(w), \tag{A-2}$$

The introduction of spatial grid size h in equation A-2 is because of the scaled wavenumber $w = kh$. This leads directly to the definition of the phase velocity v_h for the compact scheme

$$v_h = \frac{\omega_h}{|k|} = \frac{v\sqrt{w''(w)}}{w}. \tag{A-3}$$

Following Cohen (2002), we will introduce the normalized numerical phase velocity or numerical dispersion coefficient for semidiscretization of equation 1:

$$q_h = \frac{v_h}{v}. \tag{A-4}$$

This nondimensional quantity q_h measures the errors of the numerical phase velocity of the finite difference schemes. These phase velocities correspond to a monochromatic wave, i.e., a wave of a given frequency. Ideally, the operator should yield a normalized phase velocity of one for all wavenumber components.

Therefore, the normalized numerical phase velocity for 1D compact schemes can be represented by

$$q_h = \frac{\sqrt{w''(w)}}{w}, \tag{A-5}$$

where $w''(w)$ is defined in equation 10. Here, $w = kh$ is the scaled wavenumber.

The normalized numerical phase velocity for 2D compact schemes can be represented by

$$q_h = \frac{\sqrt{w''(w\cos\theta) + w''(w\sin\theta)}}{w}, \tag{A-6}$$

where $w''(w)$ is defined in equation 10. Here, $w = |k|h = \sqrt{k_x^2 + k_y^2}h$ is the 2D scaled wavenumber, θ is the angle between the propagation direction and the lateral x axis,

and the normalized numerical phase velocity for 3D compact schemes can be given by

$$q_h = \frac{\sqrt{w''(w\cos\theta\sin\varphi) + w''(w\sin\theta\sin\varphi) + w''(w\cos\varphi)}}{w}, \tag{A-7}$$

where $w''(w)$ is defined in equation 10. Here, $w = |k|h = \sqrt{k_x^2 + k_y^2 + k_z^2}h$ is the 3D scaled wavenumber. θ is the azimuth angle between the propagation direction and the lateral x axis. φ is the dip angle between the propagation direction and the vertical z axis.

For 2D and 3D problems, the finite-difference schemes generate different phase speeds (i.e., numerical anisotropy) for waves with different wavenumbers and orientations.

APPENDIX B

PREFACTORED COMPACT SCHEMES WITH TRIDIAGONAL SYSTEM

In this appendix, we will demonstrate the procedures of prefactorization of the compact schemes. For simplicity, we study a special case of equation 4. This is an approximation of the second derivative with a tridiagonal compact scheme that is given by

$$\alpha D_{i-1}^2 + D_i^2 + \alpha D_{i+1}^2 = b\frac{u_{i+2} - 2u_i + u_{i-2}}{4h^2} + a\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}. \tag{B-1}$$

Here a, b, and α are the finite difference coefficients that need to be determined. When α is nonzero, this is an implicit formulation based on Padé-type development. A family of finite difference schemes can be generated that will maintain fourth-order accuracy. In this case, we will only have a single free parameter such as α for optimization when we require fourth-order accuracy. In order to do that, the relations between the coefficients a, b, and α are derived by matching the Taylor series coefficients until fourth-order accuracy, which implies that the following relations have to be satisfied:

$$a = \frac{4}{3}(1-\alpha), \tag{B-2}$$

$$b = \frac{1}{3}(-1 + 10\alpha). \tag{B-3}$$

Different choices of these coefficients will determine the formal accuracy and the resolving power of the finite difference schemes. For example, finite difference coefficients for some special cases of equation B-1 are listed below:

a) $\alpha = 0$, this becomes standard fourth-order explicit finite-difference scheme.
b) $\alpha = 1/10$, we have $b = 0$. This is the classical Padé scheme that has been extensively used in finite difference for one-way wave equation (Claerbout, 1985). It is known to be a scheme that has

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second-order formal accuracy but has a wider wavenumber range than a standard fourth-order accurate finite difference scheme.

c) $\alpha = 2/11$, we have $a = 1/11$, and $b = 3/11$. This corresponds to a sixth-order tridiagonal finite difference scheme.

d) $\alpha = 0.20353$, then we have $a = 1.06195$, $b = 0.345126$. This is a tridiagonal compact scheme that has fourth-order of formal accuracy but optimized coefficients for a wider range of wavenumbers.

In order to construct a prefactored compact scheme for equation B-1, we will need first to obtain the numerical wavenumber of equation B-1, which is given by

$$w''(w) = \frac{\left[2a + \frac{b}{2}\right] - 2a\cos(w) - \frac{b}{2}\cos(2w)}{1 + 2\alpha\cos(w)}. \quad \text{(B-4)}$$

As a special case of equations 16 and 17, the corresponding prefactored small-stencil compact scheme for equation B-1 can be described by

$$\alpha_F D_{i+1}^{2,F} + \beta_F D_i^{2,F} = \frac{1}{h^2}[b_F u_{i+1} + c_F u_i + d_F u_{i-1}], \quad \text{(B-5)}$$

$$\beta_F D_i^{2,B} + \alpha_F D_{i-1}^{2,B} = \frac{1}{h^2}[d_F u_{i+1} + c_F u_i + b_F u_{i-1}]. \quad \text{(B-6)}$$

The real component of the numerical wavenumber of equation B-5 is given by

$$Re(w_F'') = \frac{(b_F\alpha_F + c_F\beta_F) + (b_F\beta_F + c_F\alpha_F + d_F\beta_F)\cos(w) + d_F\alpha_F\cos(2w)}{\alpha_F^2 + \beta_F^2 + 2\alpha_F\beta_F\cos(w)}. \quad \text{(B-7)}$$

A comparison of this equation with equation B-4 indicates that it is sufficient to consider a simplified form of the generic forward biased stencil to replicate this scheme's characteristics. The coefficients can be obtained by solving the following system of equations:

$$\alpha_F^2 + \beta_F^2 = 1, \quad \text{(B-8)}$$

$$\alpha_F\beta_F = \alpha, \quad \text{(B-9)}$$

$$b_F\alpha_F + c_F\beta_F = 2a + \frac{b}{2}, \quad \text{(B-10)}$$

$$b_F\beta_F + c_F\alpha_F + d_F\beta_F = -2a, \quad \text{(B-11)}$$

$$d_F\alpha_F = -\frac{b}{2}. \quad \text{(B-12)}$$

Due to the quadratic terms in equation B-8, the coefficients can not be determined uniquely by the above system alone. For this case, we choose the one that has both positive α_F and β_F where the ratio between them is minimum. The solution follows:

$$\alpha_F = \frac{\sqrt{1 - \sqrt{1 - 4\alpha^2}}}{\sqrt{2}}, \quad \text{(B-13)}$$

$$\beta_F = \frac{\sqrt{2}\alpha}{\sqrt{1 - \sqrt{1 - 4\alpha^2}}}, \quad \text{(B-14)}$$

$$b_F = \frac{\sqrt{2 - 2\sqrt{1 - 4\alpha^2}}}{8\alpha^2(-1 + 2\alpha)}\Big[b(-1 + 2\alpha)(1 + \sqrt{1 - 4\alpha^2}) + 4a\alpha(1 - 2\alpha + \sqrt{1 - 4\alpha^2})\Big], \quad \text{(B-15)}$$

$$c_F = -\frac{a\sqrt{2 - 2\sqrt{1 - 4\alpha^2}}(1 - 2\alpha + \sqrt{1 - 4\alpha^2})}{2\alpha(-1 + 2\alpha)}, \quad \text{(B-16)}$$

$$d_F = -\frac{b}{\sqrt{2 - 2\sqrt{1 - 4\alpha^2}}}. \quad \text{(B-17)}$$

Replacing the known optimized coefficients $a = 1.06195$, $b = 0.345126$, and $\alpha = 0.20353$ into the above equations, we have the prefactorization coefficients for the optimized compact scheme: $\alpha_F = 0.208085$, $\beta_F = 0.978111$, $b_F = -1.92893$, $c_F = 2.75822$, and $d_F = -0.829292$.

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